

Task 5: Exploratory Data Analysis (EDA)

Objective: Extract insights using visual and statistical exploration.

Tools: Python (Pandas, Matplotlib, Seaborn)

Deliverables: Jupyter Notebook + PDF report of findings

Hints/Mini Guide:

- Use `.describe()`, `.info()`, `.value_counts()`
- Use `sns.pairplot()`, `sns.heatmap()` for visualization
- Identify relationships and trends
- Plot histograms, boxplots, scatterplots
- Write observations for each visual
- Provide summary of findings

Outcome: Gain skill in finding patterns, trends, and anomalies.

Interview Questions:

- Which plots do you use to check correlation?
- How do you handle skewed data?
- How to detect multicollinearity?
- What are univariate, bivariate, and multivariate analyses?
- Difference between heatmap and pairplot?
- How do you summarize your insights?
- What is EDA and why is it important?

Task Submission Guidelines

- **Time Window:**
complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at
- **Self-Research Allowed:**
You are Debugfree to Yourself: explore, Google, or refer to tutorials to understand concepts and complete the task effectively. Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.
- **No Paid Tools:**
If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.
- **GitHub Submission:**
Create a new GitHub repository for each task.
Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

